

Health and Hourly Wages in the UK Labour Market

Evidence from the 2026 Q1 Quarterly Labour Force Survey

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Abstract

Using 7,372 permanent employees aged 21–60 from the UK Quarterly Labour Force Survey (Q1 2026), this paper estimates a semi-log Mincer wage equation to examine whether self-reported health status is associated with hourly wages. An extended model incorporating industry fixed effects, tenure, ethnicity, region, and work-from-home (WFH) status raises the baseline R^2 from 0.221 to 0.305. Under HC1-robust inference, workers with a limiting health condition earn 8.9% less per hour than healthy counterparts; a non-limiting condition carries no significant wage penalty at the 5% level. Structural stability tests reveal intercept differences between health groups but no slope differences, supporting the pooled specification with health dummies. A $\text{WFH} \times \text{health}$ interaction yields a joint Wald p -value of 0.64, indicating that remote work does not moderate the health wage penalty. A Heckman two-step selection correction, exploiting the presence of dependent children and marital status as exclusion restrictions, produces a highly significant inverse Mills ratio ($\hat{\lambda} = -0.446, p < 0.001$) and eliminates the estimated limiting-health penalty entirely. This suggests the OLS wage gap is driven primarily by selective employment into permanent work rather than by within-workplace wage discrimination against health-impaired workers.

1 Introduction

The determinants of wage inequality have long been central to labour economics. Education, experience, and gender feature prominently in the usual Mincer earnings equation [9], but health status has received comparatively less attention in empirical wage equations despite its obvious relevance. Poor health can reduce productivity directly, increase workplace absences, and shift workers toward lower-paying occupations or part-time contracts. At the same time, the composition of health-impaired workers in permanent employment is unlikely to be random as individuals with serious health conditions face a higher hurdle to obtain and retain a permanent job. Those who succeed may be positively selected on unobservable dimensions such as motivation or cognitive

ability [5].

This paper asks two questions: first, do permanent employees with self-reported health conditions earn systematically different hourly wages from healthy workers in the 2026 UK labour market? Second, to what extent does the estimated wage gap reflect a genuine within-employment penalty rather than a compositional artefact of who gets hired into permanent work?

We answer these questions using the Q1 2026 wave of the Quarterly Labour Force Survey (QLFS), a nationally representative household survey of approximately 40,000 working-age adults. After restricting to permanent employees aged 21–60, we estimate the baseline Mincer equation augmented with health dummies and then extend it to control for industry, firm size, tenure, ethnicity, region, and WFH status. Heteroscedasticity-robust standard errors are used after Breusch–Pagan and White tests reject homoscedasticity. A Heckman [8] two-step correction addresses the selection of health-impaired workers into permanent employment.

The principal finding is that the OLS wage penalty for limiting health conditions (-8.9%) disappears entirely once selection into employment is controlled for, reversing to a statistically insignificant positive coefficient. This implies that health-related wage inequality in permanent employment is better understood as a barrier to entry into employment than as a discriminatory pay differential within employment.

2 Data

2.1 Source and Sample

The data are drawn from the UK Quarterly Labour Force Survey, Q1 (January–March) 2026, a representative random sample of approximately 70,000 households published by the Office for National Statistics. The raw Q1 2026 wave contains around 97,000 individual observations.

The estimation sample is restricted to: (i) employees whose contract type is recorded as permanent (JOBTYP) and excludes temporary, agency and casual workers who have fundamentally different wage-setting. (ii) Individuals aged 21–60 to exclude most full-time students at the lower end and those beyond the then-current State Pension age at the upper end [6]. (iii) Observations with positive hourly pay and valid education-leaving age (EDAGE). After these restrictions the sample contains **7,372 observations**. We drop 9 observations due to missing tenure relative to the 7,381 with complete wage data.

2.2 Variable Construction

The dependent variable is the natural logarithm of gross hourly pay (HOURPAY), which transforms multiplicative wage effects into approximately additive percentage effects. Experience is proxied by

$potexp = age - edage$, the conventional Mincer approach. We include a squared term of experience ($potexp^2$) to capture the declining marginal return to experience.

Health status is coded as two binary dummies from the LIMITA variable: **limited** equals one if the respondent reports a health condition that limits the amount of paid work they can do and **nonlimited** equals one if they report a condition that does not limit paid work. The omitted category is healthy workers with no self-reported health condition.

The extended model adds firm size (**small**), four region dummies relative to London and the South East, employer tenure (midpoint-coded from EMPLN), ethnicity (**nonwhite**), WFH status (**wfh**), and twenty one-letter SIC industry dummies (reference: Section Q, Health and Social Work).

2.3 Summary Statistics

Table 1: Summary statistics — estimation sample ($n = 7,372$)

	Mean	Std. Dev.	Min	Max
Hourly wage (£)	22.7800	19.6380	0.0700	833.3000
ln(hourly wage)	2.9710	0.5500	-2.6590	6.7250
Potential experience (yrs)	22.3310	11.4280	0.0000	48.0000
Female (= 1)	0.5270	0.4990	0.0000	1.0000
Part-time (= 1)	0.1980	0.3980	0.0000	1.0000
GCSE/O-level (= 1)	0.1100	0.3130	0.0000	1.0000
A-level (= 1)	0.0730	0.2600	0.0000	1.0000
Degree or higher (= 1)	0.4870	0.5000	0.0000	1.0000
Manager/senior official (= 1)	0.3980	0.4900	0.0000	1.0000
Limiting health condition (= 1)	0.0990	0.2990	0.0000	1.0000
Non-limiting health condition (= 1)	0.2380	0.4260	0.0000	1.0000
Employer tenure (yrs)	9.0510	7.9580	0.1250	25.0000
Non-white ethnicity (= 1)	0.1360	0.3430	0.0000	1.0000
Works from home (= 1)	0.3020	0.4590	0.0000	1.0000

The sample is 52.7% female and 19.8% part-time. Nearly half (48.7%) hold a degree or higher qualification, reflecting the restriction to permanent employees in the formal sector. Around 9.9% report a limiting health condition and 23.8% a non-limiting condition; the remainder (66.4%) are healthy. The mean hourly wage is £22.78 (median implied by the log-mean of £19.57), and 30.2% of workers report working from home at least some of the time. Mean employer tenure is 9.05 years.

3 Econometric Model

3.1 Baseline Specification

The baseline wage equation follows the semi-log Mincer form:

$$\begin{aligned} \ln(hourwage_i) = & \beta_0 + \beta_1 potexp_i + \beta_2 potexp_i^2 + \beta_3 female_i + \beta_4 pt_i \\ & + \beta_5 gcse_i + \beta_6 alevel_i + \beta_7 degree_i + \beta_8 manager_i \\ & + \beta_9 limited_i + \beta_{10} nonlimited_i + u_i \end{aligned} \quad (1)$$

The log transformation means coefficients on continuous variables approximate percentage wage changes; for binary dummy variables the exact percentage effect is $100(e^{\hat{\beta}} - 1)\%$. The quadratic in experience captures the concave, hump-shaped experience–earnings profile well-established in the literature [2]. Education dummies are ordered: **gcse** and **alevel** are relative to holding no qualification; **degree** captures first and higher degrees. Healthy workers (no health condition) are the omitted reference category, so $\hat{\beta}_9$ and $\hat{\beta}_{10}$ measure the wage differential for limited and non-limited workers relative to healthy workers, conditional on all other covariates.

3.2 Extended Specification

LM tests (reported in Section 5) strongly rejected the baseline specification and motivated the addition of firm size, region, tenure, ethnicity, WFH status, and 20 industry fixed effects:

$$\begin{aligned} \ln(hourwage_i) = & \mathbf{x}_i' \beta + \gamma_1 small_i + \gamma_R' \mathbf{region}_i + \gamma_3 tenure_i \\ & + \gamma_4 nonwhite_i + \gamma_5 wfh_i + \delta' \mathbf{ind}_i + u_i \end{aligned} \quad (2)$$

where $\mathbf{x}_i$ contains the baseline regressors, $\mathbf{region}_i$ the four regional dummies (relative to London/South East), and $\mathbf{ind}_i$ the nineteen industry dummies (relative to Section Q, Health and Social Work).

4 Hypotheses

For each regressor, we will test the **Individual significance** ($H_0 : \beta_j = 0$). For us, the primary interest lies in the β_9 (limited) and β_{10} (nonlimited) variables. A rejection of the null implies a statistically meaningful wage differential for that health group after controlling for human-capital and demographic characteristics.

An **Overall significance** ($H_0 : \beta_1 = \dots = \beta_{10} = 0$) performs joint F-test on all slope coefficients. Rejection confirms that the regressors collectively explain wage variation beyond sampling error

[11].

We test **Structural stability** (H_0 : the wage equation is the same across health groups) using a Chow [3] F-test and an HC1-robust Wald test on interaction terms. Failure to reject the slopes-only restriction supports the pooled specification with health dummies; rejection would imply that the returns to experience, education, and other covariates differ across health groups.

Finally, we test for **No selection bias** ($H_0 : \lambda = 0$, where λ is the inverse Mills ratio coefficient) using a Heckman [8] two-step correction. Health-impaired workers face a higher barrier to permanent employment, so the observed wage sample may be positively selected. A significant Mills ratio would indicate that OLS estimates of β_9 and β_{10} are biased and should be corrected.

5 Diagnostic Tests

5.1 Functional Form — RESET Test

The Ramsey RESET test [10] adds powers $\hat{y}^2, \hat{y}^3, \hat{y}^4$ to the regression and tests their joint significance. Under correct specification, these terms should add no explanatory power.

Table 2: RESET test (Ramsey 1969), powers 2–4 — baseline model

	F-stat	p-value	Verdict
Baseline	4.3061	0.0048	Reject H_0

The baseline model rejects the null of correct functional form ($F = 4.31$, $p = 0.005$), indicating omitted variables or nonlinearities. The MWD test below examines whether the log-linear functional form itself is the source of misspecification; the LM tests subsequently identify specific omitted variable groups.

5.2 MacKinnon-White-Davidson Test

The MacKinnon-White-Davidson [4] test discriminates between a linear level model ($hourwage_i = X_i' \beta + u_i$) and the log-linear form used here ($\ln(hourwage_i) = X_i' \gamma + v_i$). Two auxiliary variables are constructed from the fitted values of each model:

- $z_1 = \ln(\hat{y}_{LIN}) - \hat{y}_{LOG}$: added to the log-linear model to test H_0 : the linear specification is correct.
- $z_2 = \exp(\hat{y}_{LOG}) - \hat{y}_{LIN}$: added to the linear model to test H_0 : the log-linear specification is correct.

Each null is tested via the t -statistic on the added variable.

Table 3: MacKinnon-White-Davidson test — baseline model

H_0	Added variable	t -stat	p -value	Verdict
Linear is correct	z_1 to log-linear	-2.3218	0.0203	Reject H_0
Log-linear is correct	z_2 to linear	5.3811	< 0.0001	Reject H_0

Both hypotheses are rejected at the 5% level. Rejection of the linear model ($t = -2.32$, $p = 0.020$) confirms that the log-linear form captures wage variation that the linear alternative cannot. Rejection of the log-linear model ($t = 5.38$, $p < 0.001$) is partly a consequence of the same misspecification flagged by the RESET test: with the extended model’s additional controls, this power inflates. As the MWD test does not point unambiguously toward a third alternative, we retain the semi-log specification on theoretical grounds as it is standard in the Mincer earnings literature and produces directly interpretable percentage-effect coefficients. The omitted variable groups identified by the LM tests below are the primary route to reducing the residual misspecification.

5.3 LM Tests for Omitted Variables

The LM (score) test for omitted variables regresses baseline OLS residuals on the baseline regressors plus the candidate variables, and tests the joint significance of the latter using $LM = n \cdot R_{aux}^2 \sim \chi^2(q)$, where q is the number of candidate variables.

Table 4: LM tests for omitted variable groups

Variable group	q	LM stat	p -value	Verdict
Regional dummies	4	90.05	< 0.0001	Reject H_0
Firm size	1	127.2	< 0.0001	Reject H_0
Employer tenure	1	34.44	< 0.0001	Reject H_0
Industry (SIC)	20	471.62	< 0.0001	Reject H_0
Ethnicity	1	11.79	0.0006	Reject H_0
Work from home	1	245.25	< 0.0001	Reject H_0

Every group is jointly significant at any conventional level. The largest contributions come from industry ($LM = 471.6$, 19 dummies), WFH status ($LM = 245.3$), and regional location ($LM = 213.7$). These tests motivate the extended specification in Equation 2.

5.4 Heteroscedasticity

Table 5: Heteroscedasticity tests — extended model

Test	LM stat	p-value	Verdict
Breusch–Pagan	75.46	0.0003	Reject H_0
White	686.38	< 0.0001	Reject H_0

Both the Breusch–Pagan ($LM = 75.46$, $p < 0.001$) and White [12] ($LM = 686.38$, $p \approx 0$) tests strongly reject homoscedasticity in the extended model. OLS standard errors are therefore invalid. All inference from this point forward uses HC1 heteroscedasticity-consistent standard errors.

6 Results

6.1 Extended Model Estimates

Table 6: Extended wage equation — HC1 robust standard errors

	Coeff	Std. Err.	p-value
Potential experience	+0.0185***	0.0019	0.0000
Potential experience ²	-0.0003***	0.0000	0.0000
Female	-0.1075***	0.0123	0.0000
Part-time	-0.0468***	0.0152	0.0021
GCSE/O-level	-0.0627***	0.0199	0.0016
A-level	+0.0546**	0.0224	0.0146
Degree or higher	+0.2694***	0.0126	0.0000
Manager/senior official	+0.2329***	0.0118	0.0000
Limiting health condition	-0.0888***	0.0201	0.0000
Non-limiting health condition	-0.0213*	0.0121	0.0781
Small firm	-0.1308***	0.0127	0.0000
Rest of England	-0.1675***	0.0261	0.0000
Scotland	-0.1448***	0.0321	0.0000
Wales	-0.2200***	0.0371	0.0000
Northern Ireland	-0.2577***	0.0323	0.0000
Employer tenure (yrs)	+0.0037***	0.0008	0.0000
Non-white ethnicity	-0.0746***	0.0168	0.0000
Works from home	+0.1382***	0.0138	0.0000

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Industry fixed effects (19 SIC section dummies,

reference: Section Q — Health and Social Work) included but not reported here; full estimates in Appendix B. $n = 7,372$, $R^2 = 0.305$, $\text{Adj-}R^2 = 0.301$.

Health coefficients. Workers with a limiting health condition earn 8.9% less per hour than observationally equivalent healthy workers ($\hat{\beta}_9 = -0.089$, $p < 0.001$). The exact percentage effect is $100(e^{-0.089} - 1) \approx -8.5\%$. Workers with a non-limiting condition earn 2.1% less, but this coefficient is not significant at the 5% level ($p = 0.078$), consistent with the interpretation that a health condition that does not restrict paid work has no material wage effect once other characteristics are held constant.

Human capital. The experience profile is positive and concave as expected: each additional year of potential experience adds 1.85% to wages, with the quadratic term implying a peak at $0.0185/(2 \times 0.0003) \approx 31$ years. Degree holders earn 26.9% more than workers with no qualification; A-level holders earn 5.5% more. GCSE holders earn 6.3% *less* than unqualified workers, suggesting that in this sample the GCSE group is disproportionately found in lower-paying occupations even after controlling for other observables.

Other controls. Female workers earn 10.8% less, part-time workers earn 4.7% less, managers earn 23.3% more, and workers in small firms earn 13.1% less. All regional dummies are negative relative to London and the South East, with the gradient largest for Northern Ireland (-25.8%) and Wales (-22.0%). WFH is associated with a 13.8% wage premium, consistent with WFH being concentrated in higher-skilled, higher-paying occupations [1]. Each additional year of tenure adds 0.37% to wages. Non-white workers earn 7.5% less than white workers, conditional on other characteristics.

6.2 Multicollinearity

Table 7: Variance Inflation Factors — extended model (main variables)

Variable	VIF
Potential experience	18.43
Potential experience ²	17.96
Female	1.25
Part-time	1.23
GCSE/O-level	1.23
A-level	1.17
Degree or higher	1.53
Manager/senior official	1.13
Limiting health condition	1.07
Non-limiting health condition	1.06

Small firm	1.08
Rest of England	2.85
Scotland	1.97
Wales	1.47
Northern Ireland	1.70
Employer tenure (yrs)	1.27
Non-white ethnicity	1.10
Works from home	1.20

The VIFs for `potexp` and `potexp`² are elevated (~ 18), but this is a structural consequence of including both a variable and its square in the same model and does not indicate harmful multicollinearity. The coefficients of primary interest (`limited` and `nonlimited`) have VIFs of 1.07 and below which confirms that health status is not collinear with the other regressors. All remaining variables are well below the conventional threshold of 10. Industry dummies (not shown) similarly exhibit VIFs below 2.

7 Structural Stability

A key modelling choice is whether to pool all health groups into a single equation with health dummies, or to estimate separate equations by health status. The Chow [3] test provides a formal assessment.

7.1 Classical Chow F-test

Partitioning the sample into healthy ($n = 4,815$) and non-healthy ($n = 2,557$) groups and comparing the pooled versus unrestricted RSS using the within-sample formula (same eight-variable formula in all models to ensure comparability):

$$F = \frac{(RSS_R - RSS_{UR})/k}{RSS_{UR}/(n - 2k)} = 2.65, \quad p = 0.005$$

This rejects poolability at 5%. However, given that heteroscedasticity is established (Section 5.4), the classical Chow F-statistic is invalid under heteroscedasticity and is reported here for completeness only. The HC1-robust Wald test below is the appropriate inference tool.

7.2 HC1-Robust Interaction Wald Test

Fitting the unrestricted model with full interactions between health group and all slope covariates and testing via an HC1-robust Wald test:

Table 8: HC1-robust Wald test of structural stability

Restriction	df	F-stat	5% critical	p-value	Verdict
Slopes only (8 interactions)	16	1.6125	1.6449	0.0572	Fail to reject H_0
Full (intercepts + slopes)	18	2.5411	1.6053	0.0003	Reject H_0

The slopes-only restriction fails to reject ($F = 1.61$, $p = 0.057$): the returns to experience, education, and other human-capital variables do not differ significantly across health groups. The full restriction, which also tests intercept differences, rejects ($F = 2.54$, $p < 0.001$), but this is driven by the known wage-level gap between healthy and health-impaired workers which is exactly what the health dummies in the pooled model capture.

Therefore, the pooled model with `limited` and `nonlimited` dummies is the correct specification. Estimating separate equations by health group is not warranted.

8 Extensions

8.1 Working From Home as a Moderator of the Health Wage Penalty

One motivation for including WFH in the extended model is that remote work may disproportionately benefit health-impaired workers by removing commuting costs, reducing workplace-based fatigue, and allowing flexible scheduling. If so, the health wage penalty should be smaller for WFH workers, and we would expect negative interaction coefficients δ_{lim} and δ_{nl} in:

$$\ln(w_i) = \mathbf{x}_i' \beta + \delta_{lim}(wfh_i \times limited_i) + \delta_{nl}(wfh_i \times nonlimited_i) + u_i$$

Table 9: WFH $\times$ health interaction — HC1 robust

Health group	Non-WFH coeff	WFH interaction (δ)	WFH-adjusted coeff
Limiting	-0.0966*** (-9.7%)	+0.0298	-0.0667 (-6.7%)

Non-limiting	-0.0170 (-1.7%)	-0.0134	-0.0304 (-3.0%)
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*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Joint Wald ($H_0: \delta_{lim} = \delta_{nl} = 0$):

$F = 0.4484$, $p = 0.6387$ — fail to reject H_0 .

The interaction coefficients are individually and jointly insignificant (joint Wald $F = 0.45$, $p = 0.64$). Although the point estimate suggests a slightly smaller penalty for WFH-enabled limited workers (-6.7% vs -9.7% for non-WFH), this difference is indistinguishable from sampling variation. The health wage penalty appears to be insensitive to whether the worker is able to work from home, suggesting that the gap is not primarily driven by physical workplace barriers. Although, this conclusion should be treated with caution given the relatively small number of limited WFH workers in the sample.

8.2 Heckman Selection Correction

8.2.1 Motivation

Workers with limiting health conditions are substantially less likely to hold a permanent job: the probit coefficient on `lim_s` in the first-stage selection equation is -0.30 ($p < 0.001$), implying a considerable employment probability gap. Those limited workers who are permanently employed are therefore likely a highly selected subset. They are individuals that are probably more productive, better educated, or more motivated than the average person with a limiting condition. OLS on the employed subsample conflates the wage effect of health with the positive selection effect, potentially understating the true population-level penalty.

The Heckman [8] two-step estimator corrects for this. In the first step, a probit model of permanent employment is estimated on the full working-age population (aged 21–60, $n = 39,636$). The predicted inverse Mills ratio $\hat{\lambda}_i = \phi(\hat{x}'_i\hat{\gamma})/\Phi(\hat{x}'_i\hat{\gamma})$ is then added to the wage equation as an additional regressor.

8.2.2 Exclusion Restrictions

Credible Heckman estimation requires at least one variable that affects employment probability but not (conditional on the wage regressors) wages. Two exclusion restrictions are used:

- **Dependent children under 16** (`has_dep_child`, from AOHL16): the presence of young children reduces employment probability through childcare costs and time constraints. But after controlling for gender, experience, and part-time status, there is no clear direct effect on the hourly wage of those who do work.

- **Married/cohabiting** (**married**, from MARDY6): partnership status is associated with higher employment probability (possibly reflecting financial motivation or stability). But its direct effect on wages is ambiguous once human-capital controls are included. This restriction is treated as secondary and its validity is assessed by checking whether the main results are materially sensitive to its inclusion.

8.2.3 First Stage

Table 10: First stage: probit of permanent employment (n = 39,636)

Variable	Coeff	Std. Err.	p-value
Intercept	-2.1344***	0.1189	0.0000
Age	+0.0605***	0.0061	0.0000
Age ²	-0.0007***	0.0001	0.0000
Female	+0.0076	0.0148	0.6095
Degree	+0.0995***	0.0168	0.0000
A-level	+0.0921***	0.0303	0.0024
GCSE	+0.0616**	0.0253	0.0149
Limiting condition	-0.2995***	0.0232	0.0000
Non-limiting condition	+0.0854***	0.0182	0.0000
Dependent children [†]	-0.0806***	0.0172	0.0000
Married/cohabiting [†]	+0.1386***	0.0176	0.0000

[†] *Exclusion restriction (excluded from wage equation).* $Pseudo-R^2 = 0.015$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A limiting health condition reduces the probit index by 0.30 standard deviations ($p < 0.001$), confirming that the health-employment selection is real and material. Both exclusion restrictions are significant in the expected directions: dependent children reduce employment probability (-0.081 , $p < 0.001$) and partnership status raises it ($+0.139$, $p < 0.001$). The modest pseudo- R^2 of 0.015 is typical of cross-sectional employment equations where many unobserved determinants (reservation wages, local labour demand, etc.) are unavailable.

8.2.4 Second Stage and Results

Table 11: Second stage: Heckman-augmented wage equation (HC1 robust)

	OLS coeff	Heckman coeff	Change
Variable			
Limiting condition	-0.0888***	+0.0222	+0.1109
Non-limiting condition	-0.0213*	-0.0506***	-0.0293
Inverse Mills ratio ($\hat{\lambda}$)	—	-0.4460***	—

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *HC1 standard errors do not propagate first-stage sampling error; bootstrap standard errors are reported below.*

The inverse Mills ratio is negative and highly significant ($\hat{\lambda} = -0.446$, $p < 0.001$), confirming substantial selection bias in the OLS estimates. The negative sign indicates that, conditional on observables, workers who are more likely to be selected into permanent employment earn less which is consistent with a tight labour market in which permanent employees as a group are not positively selected on unobservable productivity (high-ability individuals may prefer self-employment or contract work with higher flexibility).

Crucially, after adding the Mills ratio, the `limited` coefficient rises from -0.089 to $+0.022$ and becomes statistically insignificant ($p \approx 0.5$). The large positive correction implies that limited workers in permanent employment are strongly positively selected. They earn substantially more than a randomly drawn person with a limiting health condition would earn, to the point that, once this selection is netted out, the coefficient is indistinguishable from zero or even positive.

The `nonlimited` coefficient moves from -0.021 to -0.051 , remaining marginally insignificant. This smaller correction is consistent with the weaker employment selection for non-limiting conditions (probit coeff: $+0.085$ vs -0.299 for limiting).

8.2.5 Bootstrap Standard Errors

HC1 standard errors for the Heckman two-step estimator are inconsistent because they treat the inverse Mills ratio as a known regressor, ignoring the sampling variability introduced in the first-stage probit. A pairs bootstrap that re-estimates both stages on each draw provides asymptotically valid standard errors. The procedure resamples $B = 200$ times from the full working-age population ($n = 39,636$), re-estimates the probit, recomputes the inverse Mills ratio, and re-estimates the augmented wage equation. The standard deviation of the B coefficient vectors is used as the bootstrap standard error.

Table 12: Heckman estimates: HC1 vs bootstrap standard errors ($B = 200$)

Variable	Coeff	HC1 SE	Bootstrap SE
Limiting condition	+0.0222	0.0297	0.0304
Non-limiting condition	-0.0506***	0.0135	0.0581
Inverse Mills ratio ($\hat{\lambda}$)	-0.4460***	0.0892	0.1028

9 Discussion

The two main empirical results point in different directions regarding the mechanisms behind the health wage gap.

The structural stability tests show that the returns to human capital such as experience, education, managerial status do not differ significantly across health groups. The health wage gap is therefore an intercept shift, not a difference in the wage-experience or wage-education profile. This means that health-impaired workers are not penalised at a higher rate as they accumulate human capital but rather face a one-time level disadvantage.

The Heckman correction, however, dramatically reframes what that level disadvantage represents. The OLS estimate of -8.9% exists because the observed limited workers are positively selected; they are atypically productive relative to the average person with a limiting health condition. After controlling for selection, the penalty disappears. This suggests that in-work wage discrimination against health-impaired permanent employees is quantitatively small or absent; the real disadvantage is in the entry into permanent employment, as confirmed by the probit first stage (limiting condition: -0.30 , $p < 0.001$).

The null result on the WFH $\times$ health interaction extends this. Even within permanent employment, WFH status does not appear to be a meaningful channel through which health affects wages. This may reflect that the WFH wage premium is predominantly an occupational-sorting phenomenon (higher-paying professional roles are more WFH-compatible) rather than a barrier-removal benefit specifically for health-impaired workers.

Several limitations bear on these conclusions. First, the probit pseudo- R^2 is low (0.015), meaning many determinants of employment selection are unobserved; weak selection equations can produce unreliable Mills ratio estimates. Second, while two-step HC1 standard errors do not propagate first-stage sampling uncertainty, the $B = 200$ pairs bootstrap reported in Section 8.2 confirms that inference on the key health coefficients is robust to this correction. Bootstrap and HC1 standard errors are in close agreement for the `limited` and `mills` coefficients. Third, the exclusion restrictions, while plausible, are not bulletproof: marital status in particular may have a direct wage effect

through household bargaining or employer preferences.

10 Conclusion

This paper uses a semi-log Mincer wage equation estimated on 7,372 permanent employees from the Q1 2026 QLFS to examine the association between self-reported health status and hourly wages. The extended OLS model finds that workers with a limiting health condition earn 8.9% less per hour than otherwise equivalent healthy workers, while non-limiting conditions carry no statistically significant penalty at the 5% level.

Structural stability testing, using an HC1-robust Wald test in place of the classical Chow test which is invalid under heteroscedasticity, reveals that this gap is an intercept difference: the returns to experience and education are the same across health groups, supporting the pooled specification. Augmenting the model with WFH $\times$ health interactions produces a joint Wald p -value of 0.64, indicating that WFH does not moderate the health wage penalty.

The most substantive finding is that a Heckman selection correction, exploiting variation in dependent children and marital status as exclusion restrictions, produces a highly significant inverse Mills ratio and eliminates the estimated limiting-health penalty entirely (OLS: -8.9% ; Heckman: $+2.2\%$, insignificant). A $B = 200$ pairs bootstrap confirms that this conclusion is robust to the inconsistency of two-step HC1 standard errors. The OLS wage gap thus appears to reflect selective entry into permanent employment rather than within-workplace wage discrimination. Policy implications follow accordingly: interventions aimed at reducing the health wage gap may be more effective if targeted at removing barriers to employment entry for health-impaired workers. This can be achieved through disability employment support, flexible hiring practices, or reasonable adjustments rather than at within-workplace pay equity.

A Variable Definitions

Variable	Definition	QLFS source
<code>lnhourwage</code>	$\ln(\text{gross hourly pay, } \pounds)$	HOURLPAY
<code>potexp</code>	Age minus education-leaving age	AGE, EDAGE
<code>potexp2</code>	<code>potexp</code> squared	—
<code>female</code>	= 1 if female	SEX
<code>pt</code>	= 1 if part-time	FTPTWK
<code>gcse</code>	= 1 if highest qual. is GCSE/O-level	HIQUAL22

Variable	Definition	QLFS source
alevel	= 1 if highest qual. is A-level	HIQUAL22
degree	= 1 if holds first or higher degree	HIQUAL22
manager	= 1 if manager or senior official	MANAGER
limited	= 1 if health limits paid work	LIMITA
nonlimited	= 1 if health condition, no work limit	LIMITA
small	= 1 if employer < 25 employees [7]	MPNR02
restofeng	= 1 if region is Rest of England	GOVTOF2
scotland	= 1 if region is Scotland	COUNTRY
wales	= 1 if region is Wales	COUNTRY
ni	= 1 if region is Northern Ireland	COUNTRY
tenure	Years with current employer (midpoint-coded)	EMPLN
nonwhite	= 1 if not white ethnicity	ETHUKEUL
wfh	= 1 if works from home or home-based	HOME
sic	One-letter SIC section (20 categories)	INDS07M
has_dep_child	= 1 if has dependent child aged < 16	AOHL16
married	= 1 if married or cohabiting	MARDY6

B Industry Fixed Effects (SIC Section Dummies)

The table below reports the full set of industry coefficients from the extended HC1-robust model. The reference category is Section Q (Health and Social Work), so each coefficient measures the log-wage differential relative to that sector.

Table 13: Industry fixed effects in the extended model, HC1 robust (ref: Section Q - Health and Social Work)

SIC Section	Coeff	Std. Err.	p-value
A — Agriculture, Forestry and Fishing	-0.0001	0.0750	0.9991
B — Mining and Quarrying	+0.3099**	0.1250	0.0131
C — Manufacturing	+0.0929***	0.0226	0.0000
D — Electricity, Gas, Steam and Air Conditioning Supply	+0.2301***	0.0562	0.0000
E — Water Supply, Sewerage and Waste Management	-0.0306	0.0712	0.6677
F — Construction	+0.0855***	0.0279	0.0022
G — Wholesale and Retail Trade	-0.0763***	0.0244	0.0018
H — Transportation and Storage	+0.0828***	0.0279	0.0030
I — Accommodation and Food Service Activities	-0.1891***	0.0344	0.0000
J — Information and Communication	+0.2515***	0.0282	0.0000
K — Financial and Insurance Activities	+0.2203***	0.0289	0.0000
L — Real Estate Activities	+0.0518	0.0557	0.3523
M — Professional, Scientific and Technical Activities	+0.1468***	0.0247	0.0000
N — Administrative and Support Service Activities	+0.0081	0.0342	0.8131
O — Public Administration and Defence	+0.0812***	0.0193	0.0000
P — Education	-0.0048	0.0191	0.8018
R — Arts, Entertainment and Recreation	-0.0729**	0.0354	0.0392
S — Other Service Activities	-0.0262	0.0429	0.5413
T — Activities of Households as Employers	+0.0622	0.1062	0.5579
U — Activities of Extraterritorial Organisations	+0.1168	0.1403	0.4049

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C Full Diagnostic Output — Baseline Model

Table 14: Baseline model — OLS (for diagnostic reference only)

	Coeff	Std. Err.	p-value
Potential experience	+0.0246***	0.0021	0.0000
Potential experience ²	-0.0004***	0.0000	0.0000
Female	-0.1250***	0.0120	0.0000
Part-time	-0.1179***	0.0154	0.0000

Table 14: Baseline model — OLS (for diagnostic reference only)

	Coeff	Std. Err.	p-value
GCSE/O-level	-0.0668***	0.0199	0.0008
A-level	+0.0996***	0.0232	0.0000
Degree or higher	+0.3396***	0.0133	0.0000
Manager/senior official	+0.2388***	0.0120	0.0000
Limiting health condition	-0.1069***	0.0195	0.0000
Non-limiting health condition	-0.0136	0.0136	0.3179
—			
n = 7,381			
R ² = 0.2214			
F = 209.60 (p < 0.001)			

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